# NullahNet

**Make Mumbai's nullah trash booms accountable: clear them at the right tide, stop them leaking, pay per verified kilogram**

**Team:** ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 3:** Rising plastic pollution in coastal cities
‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

BUILT software we ran · SIMULATED digital-twin output · SYNTHETIC generated test data · SENSITIVITY assumption-driven · PLANNED not built yet

# The leak: one city, millions of kilograms a year

| What | Value | Source |
|---|---|---|
| Plastic released to the sea by the metropolitan region | an estimated 5 million kg a year | The Ocean Cleanup survey, 10 Feb 2026 |
| Damage footprint | 220 km of coastline, 152 km² of mangroves | same |
| Main pathways | "around 50 drains and nullahs" | Lokmat Times (2026), reporting the same study |
| Independent global model | Malad Creek ranked #30 of 31,819 outfalls worldwide | Meijer et al. 2021 (modelled, not measured) |
| The network that carries it | 186 outfalls; most drains built for about 25 mm/h of rain at low tide | BRIMSTOWAD, via a secondary summary |
| What one storm brings back | BMC removed 2,498.66 t of marine waste from seven beaches in 11 days (July 2026) | Mumbai Live, Jul 2026 |

# Our specific source: floating single-use flexible packaging

- Plastic is 75.5 % of 52,770 litter items counted on Juhu and Aksa beaches (CIFE study, HT 2023)
- Carry bags, films and multilayer sachets have almost no resale value. Waste-picker groups report companies ignore them (about 60% of one group's dry waste), and that sachets are sometimes **given to cement kilns for free** (Break Free From Plastic, 2023, pp. 40–41)
- The 2018 ban hasn't stopped them. In 2024 BMC fined 44,448 establishments but confiscated only 2,148 kg (Mumbai Live)
- An older BMC estimate: 500 t/day is never collected and "ends up in the storm water drains" (Citizen Matters)

**Why this source:** it is light, so the first monsoon storms flush it; it is the residue that collection misses; and it slips under booms.

# Mumbai already intercepts. The gap is accountability.

| Who | What they do | Scale (source) |
|---|---|---|
| **BMC** | Steel-net trash booms with conveyors at nullah mouths (since July 2022, on NGT directions), cleared daily by contractors | 9 booms, growing to 23; ≈ ₹1 crore each; 1.5 t/day dry, 2.5–3 t/day monsoon per boom (HT, May 2025) |
| | Two-year operations and maintenance contract | ₹14.5 crore for six rivers and nullahs (FPJ, 30 Jul 2026) |
| **The Ocean Cleanup + Plastic Fischer** | Interceptors at Malad and Trombay creeks, MCGM permits, MPCB MoU | 61 to 92 tonnes a year expected from the first systems ≈ 1.2–1.8 % of the estimated leak |
| **Plastic Fischer** (elsewhere) | Low-tech TrashBooms; kilograms verified by third-party software | 3.2 million kg, 71 booms, 8 sites in India and Indonesia |

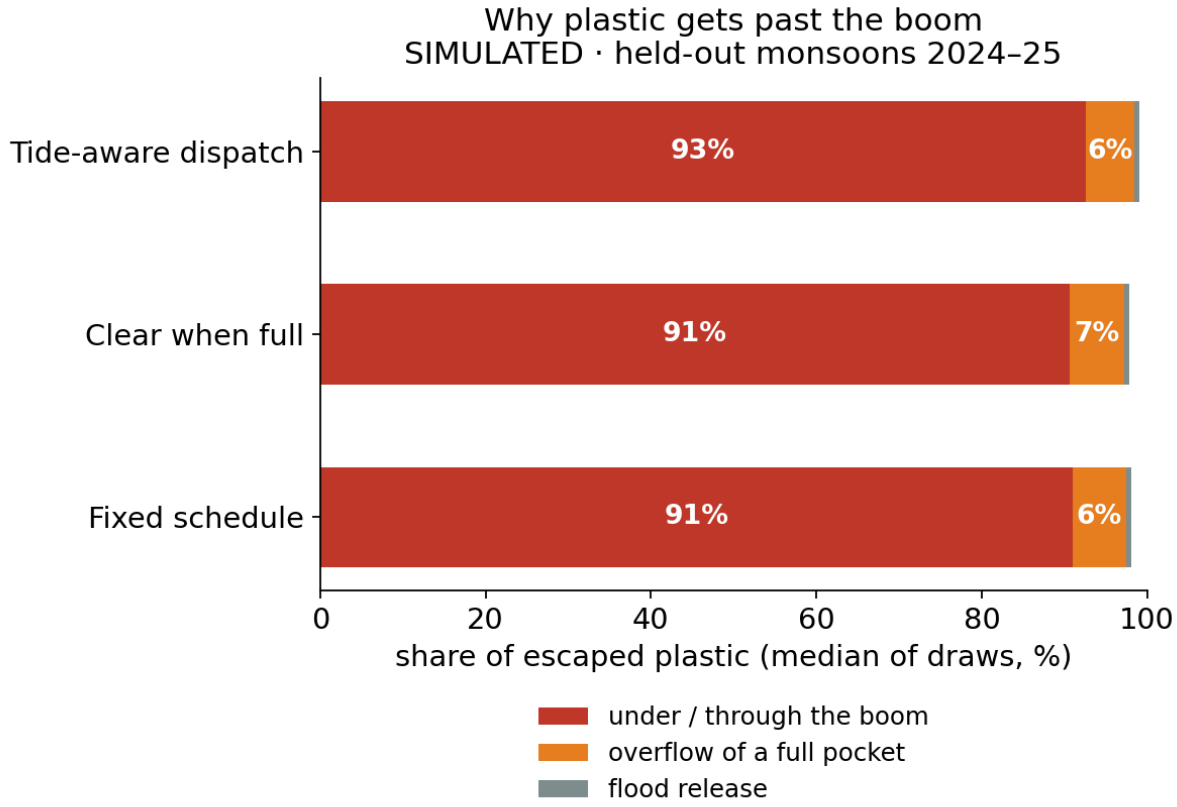
**The gap:** the booms have no sensors, clearance follows the contract rather than the tide or the rain, and payment follows claimed loads. In the Mithi desilting case, the SIT's FIR cites a ₹65.54 crore loss to BMC (allegations under investigation; The Hindu, May 2025).

# Where plastic gets past a boom SIMULATED

Digital twin of four Mumbai outfalls, held-out monsoons 2024–25, 40 Monte Carlo draws:

- **92.5%** of escaped plastic slips **under or through the boom** (interval 64.6%–100.0%), including the leak the twin assumes even in still water
- Overflow of a full pocket: 5.9% · flood release: 0.7%
- Booms lose their catch above about 0.35 m/s across the boom, whatever the skirt depth (ITOPF)
- The flood tide lifts back 23.5% of arrivals, to re-arrive on the ebb

**Levers:** angle the boom · stop the flood tide emptying the pocket · time clearance · measure




'Under the boom' includes the leak the twin assumes even in still water. Source: components/twin_sim/results/analysis.json

# NullahNet: retrofit the booms Mumbai already owns

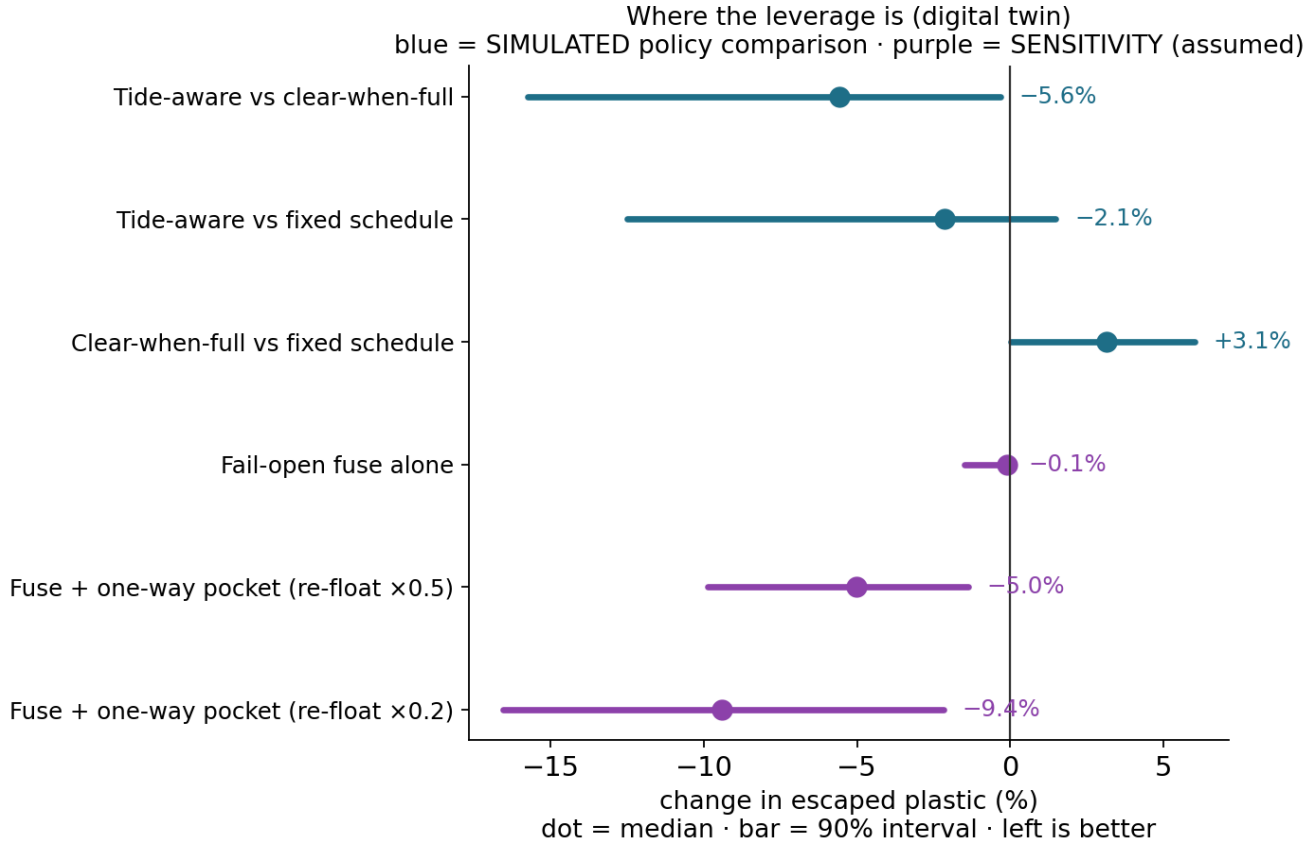
| Layer | What it is | Status |
|---|---|---|
| **1 · Fill eye** | Solar camera + ultrasonic range sensor over each boom pocket: how full, how fast it's filling | node designed and priced (PLANNED) |
| **2 · Tide-and-rain twin** | Per outfall: real ERA5 rain, archived JMA day-ahead forecasts, harmonic tide (fit RMSE 0.08 m at the Mithi) | BUILT |
| **3 · Crew dispatch** | Each morning: which boom to clear and when, timed to tide windows and the next 24 h of forecast rain | BUILT (in the twin) |
| **4 · Hardware retrofit** | Angle the boom; add a one-way pocket so the flood tide can't lift the catch out; add a **fail-open fuse** so a jammed boom never raises upstream flood levels | PLANNED (flume test) |
| **5 · Signed evidence records** | Every clearance: before/after fill, weighbridge, GPS; hash-chained and signed; paid per verified kg | BUILT |

Prior art we build on: passive floating litter traps for tidal waterways already exist (Bandalong, "wider than 2 metres, including waterways subject to tidal action"). Our contribution is the retrofit for BMC's booms, plus the operations and the evidence.

# Evidence 1 · Timing the crews SIMULATED

Held-out monsoons 2024–25; equal crew-hours (2 crews x 8 h/day, all runs); paired over 40 draws:

- **Tide-aware vs clear-when-full:** escaped plastic **–5.6%** (90% interval –15.8% to –0.3%); better in **39 of 40** draws
- **vs today's fixed schedule:** –2.1% (interval –12.5% to +1.5%): small and uncertain
- Clear-when-full alone is *worse* than the fixed schedule (+3.1%)
- Decisions use only forecasts archived before the fact




Paired over the same Monte Carlo draws, held-out 2024–25. Source: components/twin_sim/results/analysis.json

# Evidence 2 · Hardware levers, sensing, records

**Retrofit levers in the twin** SENSITIVITY (P2 dispatch, relative to today's boom):

- Fail-open fuse alone: –0.1%. It is a **flood-safety** feature, not a capture feature.
- One-way pocket that halves tidal re-floating: –5.0% (interval –9.9% to –1.4%); cut to a fifth: –9.4%. These rest on an assumed retention factor that the flume must measure.

**Detector** BUILT: off-the-shelf (COCO-pretrained, zero-shot) YOLO11n on 254 held-out floating-litter test images: mAP50 0.394, but recall 0.140 at score 0.25. Off-the-shelf models miss most river litter; training on floating-litter data is required.

**Evidence records** SYNTHETIC test data: inflated claims caught 71.6% at 4.7% false flags (a claimed-vs-sensor ratio check: 67.5%; today's slip check: 0%). Tampering detected: 100%. Precision is only 44.5% when one claim in twenty is fraudulent, so **a flag triggers a review, never an automatic penalty**.

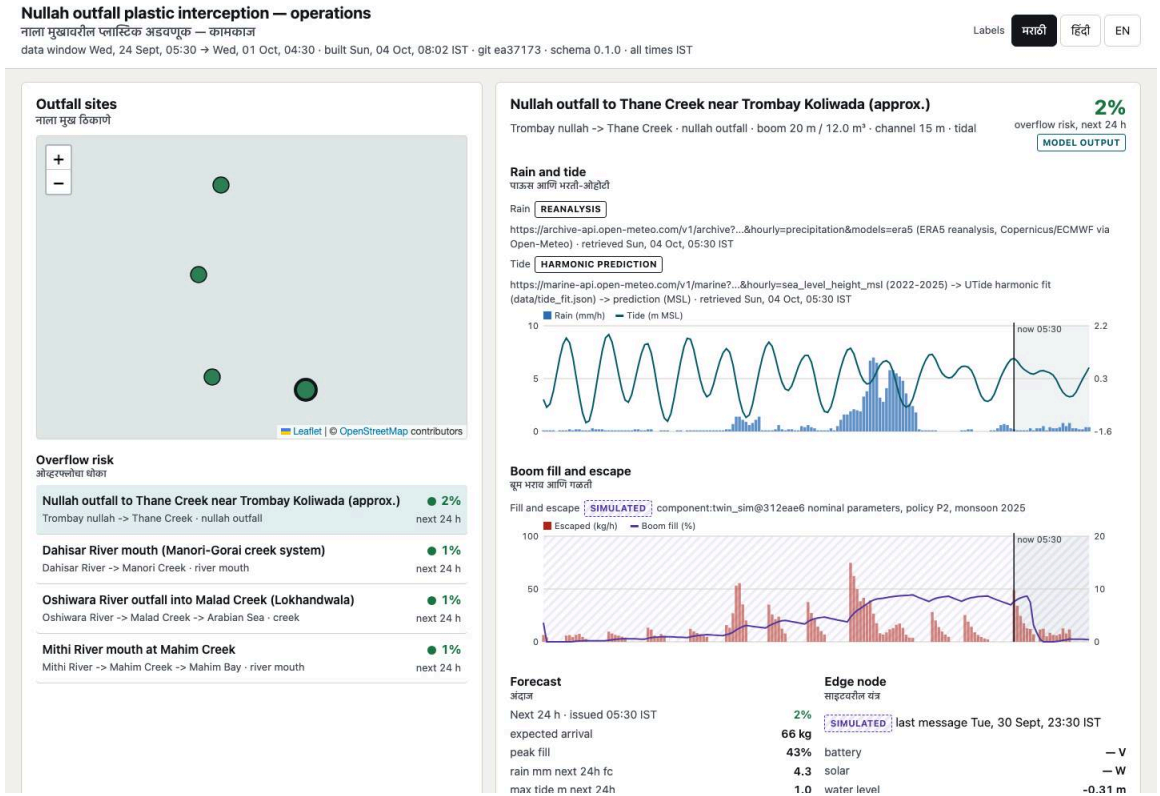
# Built for Mumbai's conditions, priced from real parts

| Challenge | Design answer |
|---|---|
| Tides reverse the flow twice a day | Twin models each site's tide; dispatch uses tide windows; the one-way pocket stops the flood tide lifting the catch |
| Monsoon surges | Fail-open fuse at a set head; the pocket sits outside the fuse path so stored catch stays |
| Salt, silt, hyacinth | HDPE floats, SS316 fittings, node on a pole above the high-water mark (traps already use HDPE and marine-grade aluminium: Bandalong) |
| Monsoon power | Sized for July: 3.79 kWh/m²/day vs 7.09 in May (NASA POWER) |
| Cost | One sensor node: ₹83,481.98 of priced parts, enclosure included, ≈ 0.8 % of one boom; boom O&M runs ≈ ₹1.2 crore per site-year (our arithmetic from the FPJ contract) |

# Who pays, the pilot we ask for, and the road to December

- **Who pays:** BMC already funds booms and drains (budget ₹80,952.56 crore for 2026-27; roads and storm-water drains take more than half). Producers under EPR: Category II covers "carry bags, plastic sachet or pouches"; low-value film can go to cement-kiln co-processing. *Whether drain-recovered plastic earns EPR certificates is not settled in the rules: we treat it as upside.* Verified-kilogram credits already sell (Plastic Fischer).
- **Hook:** BMC's ₹13,000 crore flood plan (work from October 2026) adds 8 to 10 feet mesh screens and expands water-level sensors from seven pilot sites. NullahNet is the measurement and dispatch layer for exactly such assets.
- **Pilot ask:** 2–3 small outfalls in one ward (one gated, one open), away from the Interceptor sites.
- **Roadmap:** Oct, reversed-flow flume + fuse tests → Nov, edge node on the flume (before exams) → 1 Dec, report + working prototype → Dec, live flume + dashboard demo.

# Backup · The dashboard BUILT · values SIMULATED




Real rain (ERA5) and tide (harmonic fit) with provenance badges; boom fill and escape hatched SIMULATED; Marathi and Hindi labels; runs offline.

Every number: research/fact-check.md or eval/RESULTS.md

# Backup · How the twin works and what it leaves out

- **Inputs (real):** hourly ERA5 rain; JMA day-ahead forecasts as archived; tide = UTide fit to Open-Meteo Marine sea level (RMSE 0.08 m, 99.2% of variance at the Mithi).
- **Process (assumed):** litter stock builds between storms and washes off with rain; boom capture falls with flow speed; pocket overflows; the flood tide re-floats part of the catch; crews clear.
- **Every assumed parameter** is drawn across a range in 40 Monte Carlo draws; tuning used 2019–2023 only; 2024–25 were held out.
- **Circularity:** escaped kilograms are computed by the same assumed model the policies act on. Results rank options *under stated assumptions*; field calibration (tagged-item releases at real outfalls) is the Round-2 plan.
- **Not modelled yet:** flap-gated outfalls, travel time between sites, minimum clearance duration, a perfect-foresight upper bound.

# Backup · Limits we state up front

| Claim we do not make | Why |
|---|---|
| Tonnes diverted per year | The twin's absolute kilograms rest on assumed load and capture curves |
| The one-way pocket works | Its effect is a SENSITIVITY until the reversed-flow flume measures retention |
| The camera counts all plastic | Night, rain, glare and submerged film; fill from the range sensor is the primary reading |
| Records stop fraud | Hashes make edits detectable; they cannot make inputs true; flags trigger review |
| Drain plastic earns EPR certificates | Not found in the rules; upside only |

# Backup · Sources

- The Ocean Cleanup, press releases, 10 Feb and 30 Apr 2026 · Lokmat Times (2026) · Plastic Bank river pages (Meijer et al. 2021)
- Hindustan Times: trash booms (May 2025), beach litter (2023) · Free Press Journal, 30 Jul 2026 · Mumbai Live (2026) · The Hindu, 7 May 2025 · Citizen Matters
- ITOPF TIP 3 (booms) · NASA POWER climatology · Bandalong / Storm Water Systems · Break Free From Plastic (2023)
- PWM Rules 2016 as amended (UNEP LEAP text) · OpenCity on the BMC budget 2026-27 · Pinwheel (Plastic Fischer)
- Every number with URL, quote and who opened it: `research/fact-check.md`. Every result with its reproduce command: `eval/RESULTS.md`.